



Advancing Voltage Control of Renewable Energy Resources Through a Transformerless DC/DC Converter: Enhancing Voltage Ratio and Efficiency

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Received: 30 April 2023 / Accepted: 13 September 2023 / Published online: 30 September 2023
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Abstract

In the realm of renewable energy utilization, the effective harnessing of low voltage sources necessitates the application of specialized interfaces like DC–DC power converters. Within this framework, step-up DC–DC power converters play a crucial and central role in enabling the efficient harvesting of renewable energy. This research presents an innovative DC–DC power converter with a high voltage transfer ratio, eliminating the need for a transformer. The converter utilizes switched inductor and capacitor techniques, demonstrating its unique advantages compared to traditional designs. Emphasizing low voltage switch stress as a technical priority, the proposed converter demonstrates a symbiotic relationship between high step-up voltage ratio and reduced power losses. Both continuous and discontinuous operating modes are examined to elucidate the steady-state behavior of the 250 W implemented prototype. The study provides comprehensive insights into the converter's capabilities, which are subsequently verified through meticulous experimental validation. The presented prototype exhibits impressive efficiency, reaching 97.1% owing to its ability to maintain low voltage switch stress during nominal operation. By effectively addressing the problem of low voltage renewable energy utilization, this research contributes to the advancement of sustainable power generation technologies.

Keywords High voltage ratio · High efficiency · DC–DC converters · Hybrid boosting technique

List of Symbols

C_1 – C_5	Input, interface and output power capacitors
D	Duty cycle
D_2 and D_3	Contribution of different status (duty cycles) in DCM operating mode
f_{sw}	Switching frequency

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G'_{CCM}	Voltage ratio in CCM operating mode considering circuit non-idealities
G_{CCM}	Voltage ratio in CCM operating mode
$I_{ave,Di}$	Average current of the i th power diode
i_{Ci}	Current of the i th capacitor
i_{Li}	Current of the i th inductor
I_{Lpeak}	Peak current of power inductors
I_{out}	Load current
$I_{rms,Li}$	RMS current of the i th power inductor
$I_{rms,Qi}$	RMS current of the i th power MOSFETs
L_1 and L_2	Inductors
P_{loss}	Converter total power loss
$P_{loss,D}$	Power loss in the diodes
$P_{loss,L}$	Power loss in the inductors
$P_{sw}^{Loss,Q}$	Switching power loss in the MOSFETs
$P_{cond}^{Loss,Q}$	Conduction power loss in the MOSFETs
P_{out}	Output power
Q_1 and Q_2	Power MOSFETs
r_D	Diode internal resistance
R_L	Load resistance
r_{Li}	Internal resistance of the i th inductor

